WORKED SOLUTION

STEP III 2021, Question 7

Pure • Complex Numbers in Geometry

STEP III 2021, Question 7

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2021 STEP papers (a zip file):

www.ocr.org.uk/Images/696588-2021-past-papers.zip

The key idea

When you add or subtract two points on the unit circle, $e^{i\theta} \pm e^{i\phi}$, take out the "average" factor $e^{i(\theta+\phi)/2}$. What is left is a pure cosine or a pure sine, which is what makes part (i) work. Part (i) then says that $(a+b)/(a-b)$ is purely imaginary, and in the complex plane "purely imaginary ratio" means "perpendicular". Everything after that is bookkeeping with sums of a, b, c, d.

Solution

Part (i)

Let $\alpha = \frac{1}{2}(\theta+\phi)$ and $\beta = \frac{1}{2}(\theta-\phi)$, so that $\theta = \alpha+\beta$ and $\phi = \alpha-\beta$. Then

$$e^{i\theta} + e^{i\phi} = e^{i\alpha}\left(e^{i\beta} + e^{-i\beta}\right) = 2\cos\beta\, e^{i\alpha},$$
$$e^{i\theta} - e^{i\phi} = e^{i\alpha}\left(e^{i\beta} - e^{-i\beta}\right) = 2i\sin\beta\, e^{i\alpha}.$$

The condition $\theta - \phi \neq 2n\pi$ says exactly that β is not a multiple of π, so $\sin\beta \neq 0$ and the denominator is non-zero. Dividing,

$$z = \frac{2\cos\beta}{2i\sin\beta} = -i\cot\beta = -i\cot\left(\tfrac{1}{2}(\theta-\phi)\right) = i\cot\left(\tfrac{1}{2}(\phi-\theta)\right),$$

using $1/i = -i$ and the fact that cot is odd. This is the required result.

Since $\cot(\frac{1}{2}(\phi-\theta))$ is real, z lies on the imaginary axis:

$$|z| = \left|\cot\left(\tfrac{1}{2}(\phi-\theta)\right)\right|, \qquad \arg z = \begin{cases} \frac{1}{2}\pi & \text{if } \cot(\frac{1}{2}(\phi-\theta)) > 0, \\ -\frac{1}{2}\pi & \text{if } \cot(\frac{1}{2}(\phi-\theta)) < 0. \end{cases}$$

If $\cot(\frac{1}{2}(\phi-\theta)) = 0$, that is $\phi-\theta$ is an odd multiple of π, then $z = 0$ and its argument is undefined. (This happens when the two points are diametrically opposite, so their sum is zero.)

Part (ii)

Since A and B are on the unit circle centred at the origin, write $a = e^{i\theta}$ and $b = e^{i\phi}$. They are distinct, so $\theta - \phi$ is not a multiple of 2π, and part (i) applies:

$$\frac{x}{a-b} = \frac{a+b}{a-b} = ik, \qquad k = \cot\Big(\tfrac{1}{2}(\phi - \theta)\Big) \in \mathbb{R}.$$

Because $x = a + b \neq 0$, we have $k \neq 0$. So $x = ik\,(a-b)$: the vector $\overrightarrow{OX}$ (represented by x) is the vector $\overrightarrow{BA}$ (represented by $a-b$) rotated through a right angle and scaled by the non-zero real number k. Equivalently, $\arg x - \arg(a-b) = \pm\frac{1}{2}\pi$. Hence OX is perpendicular to AB.

Ask yourself Why do we need $a + b \neq 0$?

If $a + b = 0$ then X is the origin itself. There is no line OX, so there is nothing to be perpendicular. In the algebra this is the case $k = 0$, where the argument of x is undefined.

Part (iii)

With $h = a + b + c$ we have

$$h - a = b + c.$$

The complex number $h - a$ represents the vector $\overrightarrow{AH}$. There are two cases.

- If $b + c = 0$, then $h - a = 0$, so $h = a$.
- If $b + c \neq 0$, apply part (ii) to the distinct points B and C on the unit circle. It says that the vector represented by $b + c$ is perpendicular to BC. That vector is $\overrightarrow{AH}$, so AH is perpendicular to BC. (Here $H \neq A$, so AH is a genuine line.)

So either $h = a$ or $AH \perp BC$, as required. By symmetry the same holds for BH and CA, and for CH and AB, so H does lie on the altitudes. The case $h = a$ is when BC is a diameter: the triangle has a right angle at A, and its orthocentre is the vertex A.

Part (iv)

By the definition of the orthocentre,

$$p = a + b + c, \qquad q = b + c + d, \qquad r = c + d + a, \qquad s = d + a + b.$$

The midpoint of AQ is

$$\tfrac{1}{2}(a + q) = \tfrac{1}{2}(a + b + c + d).$$

This expression is symmetric in a, b, c, d, so it is also the midpoint of BR, of CS and of DP:

$$\tfrac{1}{2}(b + r) = \tfrac{1}{2}(c + s) = \tfrac{1}{2}(d + p) = \tfrac{1}{2}(a + b + c + d).$$

Let M be the point represented by $m = \frac{1}{2}(a + b + c + d)$. Then

$$q = 2m - a, \qquad r = 2m - b, \qquad s = 2m - c, \qquad p = 2m - d.$$

So the single map

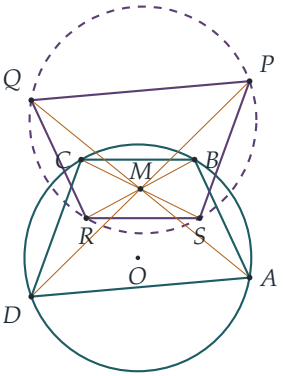
$$T : w \longmapsto 2m - w$$

sends $A \mapsto Q$, $B \mapsto R$, $C \mapsto S$ and $D \mapsto P$. It therefore maps the quadrilateral $ABCD$ onto $QRSP$, vertex by vertex.

To describe T, write it as $T(w) - m = -(w - m)$. The displacement from M is reversed, which is multiplication by $e^{i\pi}$ about M. So

T is a rotation through 180° *(a half-turn) about the point M with complex number* $\frac{1}{2}(a + b + c + d)$.

Equivalently, T is reflection in the point M. In particular $QRSP$ is congruent to $ABCD$ and, since a rotation preserves orientation, Q, R, S, P also run anticlockwise. The four orthocentres lie on the unit circle centred at the point with complex number $a + b + c + d$, the image of the circle through A, B, C, D.



Checks

- Part (i): with $\theta = 0$, $\phi = \frac{1}{2}\pi$, we get $z = (1 + \mathrm{i})/(1 - \mathrm{i}) = \mathrm{i}$, and $\mathrm{i}\cot\frac{1}{4}\pi = \mathrm{i}$. So $|z| = 1$ and $\arg z = \frac{1}{2}\pi$.
- Part (iv): take $a = 1$, $b = \mathrm{i}$, $c = -1$, $d = \frac{3}{5} - \frac{4}{5}\mathrm{i}$. Then $m = 0.3 + 0.1\mathrm{i}$, and $q = b + c + d = -0.4 + 0.2\mathrm{i}$, which equals $2m - a$ as claimed. For a square $(a, b, c, d = 1, \mathrm{i}, -1, -\mathrm{i})$ we get $m = 0$, and Q, R, S, P are just C, D, A, B: the half-turn is about O.
- All four parts were also confirmed by computer on thousands of random points on the circle.

Common slips

- Giving $\arg z = \frac{1}{2}\pi$ only. The sign of the cotangent decides between $\pm\frac{1}{2}\pi$, and $z = 0$ has no argument.
- In part (iv), matching the vertices in the wrong order. A goes to Q, not to P: the orthocentre opposite each vertex is the one of the triangle that *omits* it.

- Describing the map as "a rotation" without giving the centre and the angle, or as "an enlargement of scale factor -1" without giving the centre. Either description is fine, but it needs both pieces of information.

Beyond STEP

The neat formula $h = a + b + c$ (with the circumcentre at the origin) is the start of a whole style of complex-number geometry. For instance, the centroid is $\frac{1}{3}(a + b + c)$, so O, the centroid and H lie on a line, the Euler line. Part (iv) is a small piece of the same story for cyclic quadrilaterals. Tristan Needham's *Visual Complex Analysis* treats complex multiplication as rotation and scaling throughout, and is a superb next step.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.